Ghana Life: Master Plan

Updated Oct 9, 2026 (revision 2: quality tiers, risk register, week-one sprint)

Summary

Ghana Life is a free browser game where people live a Ghanaian life in a realistic virtual Ghana. Accra is the default home, and the whole country can be explored. Players meet in real time, talk by text and voice, go to school or learn a trade, work, run businesses, marry another player, vote, and travel. It runs from a link, with no download.

Why it can win. Lagos Life showed that a free, shareable, locally flavoured life world can reach millions of players in days. We go further with a real-looking map of the country, live voice, Ghana's school and national service path, marriage between two real players, elections, and a business tool that lets local sellers promote themselves for less.

Decisions already made

- Browser first, installable later, never required.
- Accra is the default, and the whole of Ghana is on the map.
- A real-map engine built from open data, instead of hand-drawn maps.
- Moving traffic is simulated on real roads.
- Voice is optional: private rooms for friends come first (Phase 1), and nearby voice on the open map follows later.
- The NSS allowance follows the real figure.
- Banner prices sit below Lagos Life's reported rates.
- Everything is 18+, and crime and gang play is fictional and consensual.
- Three quality tiers (Light, Standard, Ultra) are picked automatically, so weak phones run smoothly and strong devices get the full look.
- People and vehicles are drawn with a light 2D engine (PixiJS) above the map. 3D is loaded only on Ultra.
- Week one builds a test version that proves speed on cheap phones. It is not the full game.
- Players are anonymous by default and share more about themselves only by mutual choice, step by step (the Reveal Ladder).
- Public launch leads with three signature features: the alive map, the Reveal Ladder with private voice, and the Ghana culture calendar. Everything else waits.
- Scale by design: static-first files, simulation on each player's phone, many small rooms and admission control, so growth is gradual and overload becomes a queue, not a crash.

Version 1 (first playable). A real-looking Accra map, a walking avatar, live chat, a few moving cars, need bars, one job and rent. Everything else follows in phases (see the roadmap).

Honest expectations. Lagos Life's first four days were an exceptional case. Treat revenue as a result of a good game and an audience, not as something we can promise.

No lag, honestly. Nobody can promise zero lag for every player on every phone. What we can promise is a game that is measured against a performance budget, that draws less when a phone struggles, and that is tested on a cheap Android before anyone else sees it.

Scale, honestly. One million registered users is a design target, not a promise. The plan is free while small and cheap in proportion when large. Growth is controlled by invite waves, a waiting line and a spending limit, so overload becomes a queue, not a crash.

How to use this document with Claude

This is the single source of truth for the project. To build with it, give Claude the whole document, then work one task at a time from the "Build brief for Claude" section near the end. Each task there has a prompt you can paste, and a "done when" test that must pass before moving on. The rest of the document explains the design and the reasons behind it. Where this document says Claude must or must not do something, treat it as a rule.

One honest note on scope: the goal is the same kinds of features as Lagos Life, built as our own game with our own names, text, art and code. It is not an exact copy, and it must not become one, for legal reasons explained in the legal section.

Read first. The sections "Quality tiers and performance", "Look and feel", "Risk register" and "Week-one sprint" were added in revision 2. If they disagree with older text, they win. Claude must follow the performance budget on every task.

Zero-dollar plan

The whole first version must cost nothing. No paid plans, no payment card, no purchased domain, no paid services and no payment processing inside the game. Before adding any tool, Claude must check that it works with no payment method.

Part	Zero-dollar choice	Free limit	At the limit
Game files and map file	Cloudflare Pages	Unlimited static requests and bandwidth, 500 builds a month. Maximum 25 MiB per file and 20,000 files	Cut the map into several smaller files, each under 25 MiB. If that is not enough, use GitHub Pages as a backup
Live rooms	Cloudflare Workers with Durable Objects (free plan, built-in storage)	100,000 requests a day, 13,000 GB-seconds of compute a day, 100,000 stored-row writes a day, 5 GB storage. Incoming WebSocket messages	Calls fail until the daily reset at midnight UTC, which is Ghana time

Part	Zero-dollar choice	Free limit	At the limit
		count as 1 request per 20 messages	
Voice	Cloudflare Realtime	First 1,000 GB a month, no card needed to start	Cap voice use in code and keep a kill switch
Map software	MapLibre GL JS and a Protomaps basemap cut for Accra	No licence fee	None
Accounts	Nickname and passcode stored in Durable Object storage, no email	Same daily row limit	Guests can still browse
Moderation	Word list in code, a report button, and you reviewing reports	None	None
Payments	None inside the game at launch	None	Banners are placed by you after an agreement outside the game
Code and coding help	GitHub free, Claude's current plan	Message limits on Claude's free plan	Build slower, one task at a time
Art and sound	Your own, or packs licensed for free commercial use	Check each licence	Use fewer assets

What the zero-dollar plan can handle (estimate, to be load tested). The binding limit is the 100,000 requests a day. At 1 request per 20 incoming messages that is about 2,000,000 messages a day. If each active player sends about one message every 4 seconds, that is 900 an hour, so the plan supports roughly 2,200 player-hours a day, for example about 500 players for around 4 hours each. This is an estimate based on published limits and my assumptions, so measure it.

Design rules that keep it inside the limit

1. Click-to-move, not streaming. A player sends one message when they choose a destination, and others animate the walk. Why: it cuts messages by more than 90 percent compared with sending positions many times a second.
2. Server messages out to nearby players are free, so broadcast results freely, but keep incoming messages few. Why: only incoming messages count.
3. Send chat only when typed, and a heartbeat no more than once a minute.
4. Keep positions in memory, and write to storage only for important events such as jobs, wallet changes and rent. Why: stored-row writes are limited to 100,000 a day.
5. Use the WebSocket hibernation API so idle rooms stop using compute.

6. Count usage in code. At about 90 percent of the day's limit, show "the city is resting until midnight" to new arrivals. Why: a friendly message is better than random failures.

What zero dollars cannot do. It cannot give unlimited players, a satellite look (needs a card), email password reset, or taking real payments in the game. Why: each of those needs a paid or card-backed service. They are later upgrades paid for by income.

No strong computer needed. Once deployed, nothing in this plan runs on your own computer. The game files, the map files and the live rooms all sit on Cloudflare's free servers, so players never connect to your laptop. "Self-host" in this plan means putting our own map file on Cloudflare Pages, not running a server at home. Your computer is only used to write code and deploy, so a modest laptop is enough (about 8 GB of memory is comfortable, 4 GB is tight). If it is too weak, use GitHub Codespaces, a free coding environment in the browser (check its free monthly hours and that no card is needed). The tool that cuts the map is expected to download only the Accra region, not the whole world, and T2 confirms this.

Alternatives considered

Option	Why it is not the default
Firebase Realtime Database, free Spark plan	Limited to 100 simultaneous connections
Supabase free plan	200 live connections, and projects pause after 7 days of no use
Oracle Cloud Always Free server	Needs a card to sign up, capacity is hard to get, and its free ARM allowance was reported halved from 18 August 2026
Mapbox satellite	Needs a card on file
Running the server on your own computer	Free but must stay on all day

Sources: Cloudflare plans, Cloudflare Durable Objects pricing, Cloudflare Realtime pricing, Firebase Realtime Database limits, Supabase Realtime limits

The world

A real map, no drawing. The game world is the real map of Ghana, built from open data and shown in the browser. Players, vehicles and places sit on top of real streets, so Accra looks and feels like Accra. Interiors such as a chop bar, a club, a hospital or a game centre open as simple illustrated scenes. Why: you get a realistic country with almost no hand-drawing, and travel is just moving the map.

Two looks, one engine

Look	What it is	Cost and risk	When
Stylised map	The real streets and buildings in our own bright colours, from	Free to host yourself, light on mobile data	Version 1

Look	What it is	Cost and risk	When
	OpenStreetMap through Protomaps and MapLibre		
Satellite view	Real aerial photos under the same game	Mapbox offers satellite and free 50,000 map loads a month, then about 5 dollars per 1,000, and needs a card. Heavy on data. Check its terms for use in a game	Later, as an optional switch

Imagery to avoid or check. Do not use Google satellite imagery in the game: Google says satellite view falls under Google Earth's commercial restrictions and bans tracing roads from it. OpenAerialMap offers free imagery, but coverage is patchy and licences vary by image (many are CC-BY, some are not), so check each one. Why: licence problems are expensive to fix after launch.

Moving traffic. Real live traffic data is not needed. Cars, taxis, trotros, tricycles, motorbikes, bicycles and pedestrians are simulated along the real road network. Everyone sees the same traffic because it runs from a shared clock and seed, so it costs no network data. Rush hour, rain and power cuts change how it moves. Why: it looks alive, and it is free to run.

Light mode. A data-saving mode with the stylised map, fewer vehicles and no satellite. Why: many Ghanaian players are on limited mobile data. The three tiers are set out in the next section, "Quality tiers and performance".

Places. About 100 hero landmarks get illustrated interiors, and thousands of other places come from open data with generic names. Public landmarks use real names, and private businesses use generic or invented names until the owner claims the place. Examples: the presidency building, hospitals, universities, the airport, Makola and Kantamanto markets, Oxford Street, Labadi Beach, the stadium, game centres, nightlife, churches and mosques. Why: it covers popular and unpopular spots without implying a business has joined.

Travel across Ghana. The same map covers all 16 regions, so players travel by trotro, bus, taxi, ride app or a domestic flight, which costs Sika and time. Detail grows region by region: Greater Accra first, then Kumasi, Cape Coast, Tamale and others as players vote.

Characters. A layered creator with a wide range of skin tones, hairstyles such as braids, locs, short cuts and headwraps, and clothes from uniforms to kente. Start semi-realistic and improve later. Why: players want to see themselves, and heavy realism is slow to produce. The method is set out in the section "Characters and avatars".

Quality tiers and performance

The idea. One game, three quality tiers. On first load the game checks the device and runs a five-second frame-rate test, then picks Light, Standard or Ultra. The player can switch in settings, and the choice is remembered. Why: weak phones stay smooth, strong devices get the full show, and we build one game, not two.

Where lag really comes from. In map games, most lag comes from the phone drawing too much, then from slow or unstable networks, and last from the server. So the first fixes are drawing less and sending less.

The three tiers

Tier	Who gets it	What is drawn	Target
Light	Older or low-memory Android phones, slow or data-saver connections	Flat stylised map, no raised buildings, up to about 15 cars, about 20 avatars nearby, planes as simple sprites, no weather particles, flights skip the cinematic	30 frames per second, smallest download
Standard	Most mid-range phones and laptops	Stylised map with simple shadows, buildings raised near the player only, about 40 cars, clouds, day and night tint, ambient planes, a short flight cinematic	45 or more frames per second
Ultra	Recent phones, laptops and desktops with a good graphics chip	Tilted camera, 3D buildings, about 100 cars, a 3D plane model, rain and night lights, dumsor blackout waves, full flight cinematic	60 frames per second

All numbers here are starting targets. Test 1 in the week-one sprint decides the real ones.

Automatic downgrade. If the frame rate stays below the tier's target for about 3 seconds, the game removes detail one step at a time: fewer cars, then effects, then raised buildings, then it moves to the next tier down. It never upgrades by itself during play, it offers a button instead. Why: lag is fixed by drawing less, not by waiting.

Engines considered

Option	Verdict	Why
MapLibre GL JS	Use	Draws the real map in WebGL, is free, works with PMTiles, and has tilted cameras and raised buildings built in
PixiJS	Use, confirm in Test 1	Fast 2D drawing of many small sprites, free and open source. Good for avatars, cars, shadows and effects drawn above the map

Option	Verdict	Why
three.js	Use on Ultra only	Needed only for the 3D plane and special effects, so Light and Standard never download it
Phaser	Not now	A good 2D game engine, but it usually wants to own the whole screen, which clashes with a real map as the base
Unity, Godot or Unreal on the web	No	Web builds are usually large and heavy on weak phones, and they are not made for streaming a real map
deck.gl	Backup	Strong at drawing many moving points on a map, but a bigger library. Keep it as a fallback if sprites turn out too slow

Why not a single big game engine: a Ghanaian player on mobile data will leave if the first load is slow. Small libraries, loaded only when needed, keep the first load short.

How we keep it light

- Draw only what is on screen, and only the nearest players and cars.
- Use one shared sprite sheet for avatars and vehicles, and reuse sprite objects instead of creating new ones.
- Run traffic and plane paths in a Web Worker, away from the drawing code, so the screen stays smooth.
- Cap pixel density (about 1.5 on Light, 2 on the others). Why: phones draw many more pixels for little visible gain.
- Cache the map and game files on the phone with a service worker, so the second visit is fast and costs no data.
- Compress pictures (WebP) and use short audio clips. Load voice, 3D and flights only when first needed, never in Light.
- Cut the map into several small files. Each file must stay under 25 MiB, which is the Cloudflare Pages limit per file (confirmed in Cloudflare's documentation).
- Animate other players between messages, so a slow network looks smooth instead of jumpy.
- Show a loading screen with real progress and a data-saver switch that tells players roughly how much data they use.

Performance budget

Measure	Light	Standard	Ultra
Game download before the map shows	Under 3 MB	Under 8 MB	Under 20 MB
Seconds until the map shows, on 4G	Under 5	Under 5	Under 8
Frames per second	30 or more	45 or more	60

Measure	Light	Standard	Ultra
Cars drawn at once	Up to 15	Up to 40	Up to 100
Avatars drawn at once	Up to 20	Up to 40	Up to 80
Messages sent per player	Under 10 a minute on average	Under 10 a minute on average	Under 10 a minute on average

These are targets, not measured facts. Measure them on real phones with Chrome remote debugging, Lighthouse and our own frame-rate overlay.

Official test phone. Pick the cheapest Android phone you or a friend has (about 2 to 3 GB of memory). If the game is smooth there, it will be smooth on most phones. Test also on one mid-range phone and one laptop, and open the link inside the WhatsApp browser.

Look and feel: realistic, alive, exciting

Honest limits. At zero dollars, "realistic" means a real map of Ghana in our own beautiful style, not photographs. The city feels real because of motion, light and sound, not because of satellite pictures. Do not promise photo-real graphics, a perfect no-lag game or unlimited players. Promise this: smooth on cheap phones, beautiful on good ones.

What makes it feel real

- Real streets, rivers, lagoons, beaches and parks from open data, coloured in our own palette.
- Buildings raised to their real height where the data has it, and a standard height where it does not. Map data is patchy in parts of Ghana, so spot-check each district.
- Day and night that follow Ghana time, with street lights, shop lights and car lights switching on at dusk.
- Weather: harmattan haze in the dry season, and rain with wet roads and slower traffic in the wet season.
- Dumsor: a blackout that rolls across districts at night, with generators humming in some places. Why: it is local, funny and looks striking.
- Traffic with personality: trotros that stop and load, taxis that honk, jams at known bottlenecks, and tricycles weaving through.
- Sound by area: market chatter, traffic, beach waves, radio music, and church and mosque ambience handled with respect.

Planes you can see. Ambient planes fly over Accra on approach to and from Kotoka, run from the same shared clock and seed as the traffic, so every player sees the same plane in the same place at no network cost. The path is an approximate approach line, not real flight data. Light and Standard show a small sprite with a shadow, and Ultra shows a 3D model with lights at night. Why: planes overhead make the city feel alive long before the booked-flight cinematic exists in Phase 4.

The first 60 seconds, ranked

Wow moment	Impact	Difficulty	Tiers
Opening swoop: the camera flies from all of Ghana down to the player's spot in Accra	High	Easy	All (shorter in Light)
Alive roads: cars, taxis and trotros moving, with horn sounds	High	Medium	All
A plane crosses the sky overhead	High	Medium	All (3D on Ultra)
Other real players walk past, wave or chat	High	Medium	All
Dusk falls and the lights come on	High	Easy	Standard, Ultra
Rain rolls in and traffic slows	Medium	Medium	Standard, Ultra
A dumsor blackout wave crosses the district	Medium	Medium	Standard, Ultra
A character creator with real Ghanaian looks	High	Hard	All (after week one)

Build order for wow: the first four in week one if possible, then dusk, rain and dumsor, then the character creator.

The first 60 seconds

The goal. In one minute a new player should feel "this is Ghana, it is alive, and someone is here". Every step below must also work in Light, and the first load must stay inside the performance budget.

Seconds	What the player sees and hears	Notes
0 to 5	A kente-inspired loading screen with a real progress bar and the word "Akwaaba" (welcome)	Tiny download, no 3D
5 to 12	The camera swoops from all of Ghana down to Accra, with soft city sound rising	A map camera move, cheap on every tier
12 to 25	Quick start: pick male or female, a skin tone and a hairstyle, and get a day-name nickname such as "Kofi of the Lagoon". A skip button is always there	Max 15 seconds, flat preview in Light
25 to 35	You land at a beautiful landmark. Cars, trotros and tricycles move, a plane crosses the sky, and a trotro mate shouts a stop	Light and weather follow Ghana time, so it must look good at any hour, including a night with city lights
35 to 45	Tap to walk. Another player waves, and an "Agoo" button lets you knock and greet them	One tap to say hello is the social hook
45 to 60	Your phone buzzes: a friendly guide (a fictional "Auntie Ama") sends your first small gig and mentions today's event from the culture calendar	Shows that the world has things to do and a calendar

Measure it: time to the first walk, and how many new players are still playing after 5 minutes. These two numbers say whether the first minute works.

Characters and avatars

The goal. Players pick male or female, then customise skin tone, hair, face and clothes. The character looks beautiful and Ghanaian, and stays light enough for phones. Why: players look at their own character more than anything else, so it must feel good.

The method: two base bodies, many swappable parts

- Two base bodies, male and female, each with one shared skeleton, so every animation works on both.
- Skin tone is a set of 8 to 10 preset tones applied through the material colour, not 8 to 10 separate models.
- Hair, clothes, shoes and accessories are separate pieces fitted to the same skeleton, swapped in and out.
- Body parts hidden under clothes are switched off, so nothing pokes through.
- Later: face and body sliders, and a few more body builds.

Where each tier uses what

Tier	Characters
Light	Flat sprites, never real 3D
Standard	Sprites with shadows, or low-poly 3D for the nearest few players if Test 1b allows it
Ultra	Real low-poly 3D for the nearest 10 or so players, sprites for the rest
Close-up scenes	Full 3D on every tier (character creator, wedding, interiors, flight cabin), because only a few characters are on screen

Budget per character (starting targets, to be measured). About 3,000 to 5,000 triangles for a body, 500 to 1,500 for each hair or clothing piece, one compressed 1024-pixel texture, and under about 2 MB per base body including animations. Every piece shares one skeleton and one set of animations.

How to make the bodies

- Step 1. Make reference pictures: front, side and back views of each base person, in a neutral pose with plain fitted base clothing, using an image AI. Keep the same face and body in every view.
- Step 2. Turn the pictures into a 3D model with an image-to-3D tool. A multi-view mode gives a better shape. Compare Hunyuan 3D (reported to be free), Meshy, Tripo and Rodin, and check each tool's licence for commercial use.
- Step 3. Clean it up in Blender (free): reduce triangles, fix the shape, make clean textures and stand the model in an A-pose.
- Step 4. Add the skeleton with Mixamo (free, needs an Adobe account) or the tool's auto-rig. Test walk, idle, run, wave and sit.

- Step 5. Make hair and clothes as separate pieces fitted to the same skeleton. This is the hardest step and needs Blender skill or a 3D artist.
- Step 6. Export compressed GLB files, load them in the browser and test on the cheap phone.

Prompt for the base body pictures (edit and reuse)

A full-body character reference sheet of an adult Ghanaian [man or woman], neutral A-pose with arms slightly away from the body, plain fitted grey base clothing, bare feet, even studio lighting, plain light background. Front view, side view and back view of the same person, realistic proportions, stylised semi-realistic game character, no logos, no text.

Honest limits. AI shapes usually need cleanup, and hands and faces are often the weak spot. Fitting hair and clothes onto a shared skeleton is real 3D work. With two bodies, players of one gender share a body shape, so characters will feel similar until sliders and extra builds arrive. Check early that braids, locs, twists and headwraps come out well, because AI models often get them wrong. A freelance 3D artist for a few days may be worth paying later to finish the final bodies and the first set of hair and clothes. That is optional, and week one stays at zero dollars. Never use a real person's photo or likeness for a base body without permission.

Test 1b: character speed. Load 10, 20 and 30 animated low-poly 3D characters with the map on the cheap Android phone and note the frame rate. If 10 characters hold about 30 frames per second, Standard and Ultra may show real 3D for the nearest few. If not, real 3D stays in close-up scenes only.

Feature checklist: everything Lagos Life has, and our better version

The left column comes from the unofficial Lagos Life guide (figures checked in-game on 5 October 2026, so they may change). The middle and right columns are our design. We borrow the idea of each system, not its names, text or art.

Lagos Life has	Ghana Life version	Why ours is better
A birth lottery with three starts and no reroll	Five starts (compound child, returnee, graduate, market child, landlord's heir), picked or drawn	More choice, and Ghana's own starting points
Six needs: hunger, energy, hygiene, bladder, fun, social	Energy, food, hygiene, mood and Vibe, with social, health, skill and peace added later	Vibe is a new community stat
33 career tracks with five ranks each, skills, shifts and promotion by performance	Ghana careers: teaching, nursing, medicine, law, banking, civil service, tech, trading, tailoring, hairdressing, chop bar, trotro and ride driving, radio, music, football, farming, real estate, aviation, and more	Each shows a requirements card, and school and NSS feed into them

Lagos Life has	Ghana Life version	Why ours is better
Side gigs such as helping at a food spot, carrying loads, freelancing, reselling, open mic, comedy, photoshoots and dance battles	Gigs such as chop bar help, market runner, WhatsApp hustle, thrift resale, tutoring, mobile money agent, open mic, comedy night, Labadi photoshoot, dance battle	Gigs link to real Ghanaian hustles
Businesses that pay out each evening and can lose on a bad day, with 60 percent resale	Kiosks, a waakye stand, a chop bar, a salon, a barbershop, a car wash, a trotro, an event centre, a club share	Players can hire other players as staff
Land that grows weekly, with a risk of land guards demanding fees	Plots around Accra and beyond, with a real land-guard risk and a paperwork mini game	A real Ghana issue turned into gameplay
15 homes with weekly rent due on Saturday, arrears and eviction	Rooms from a hostel to a mansion, with weekly rent or a discounted rent advance	The advance option matches how rent works in Ghana
Power cuts, with generator and solar choices	Dumsor, with a generator, solar and power banks	Local and relatable
A weekly player-run governor election with policies like a transport subsidy	A ladder from neighbourhood leader to president, fictional parties, policies, term limits and a recall vote	Higher goals, and safer against abuse
Billboards, lagoon plots, music in clubs and brand apps on the phone	Banners, claim-your-place, music slots, brand apps, jobs board, property listings	Lower prices and more buyer types
28 named places across 8 areas, with opening hours, prices and activities	Thousands of imported places, about 100 illustrated landmarks, real opening hours	The whole of Ghana, not one city
Hospital, chapel, mosque, radio house, university, markets, police, court, hotel, clubs, beach, mall	The same kinds of places, with Ghanaian versions of each	Same depth, local flavour
Police, bail, fines and jail	Police, courts and jail as consequences for crime play	A clear cost for the bad-boy path

Lagos Life has	Ghana Life version	Why ours is better
A daily clue hunt for a cash prize	A daily "treasure of the day" with Ghana trivia, kente and local clues	Teaches culture while it entertains
Gifts of money between players with a small fee	A mobile-money style wallet with a small fee and scam warnings	Familiar to every Ghanaian player
Danfo buses and other rides between places	Walk, ride app, taxi, trotro, tricycle, own car, intercity bus and domestic flights, with players as drivers	Player-run transport and flights
An aviation career at the airport and airport staff jobs, with flights listed as coming soon	A full airport at Kotoka: flights between regions, cabin crew, pilots and ground staff, with a cinematic flight view	Planes that actually fly
A casino	No casino. Skill-based game centres and arcades instead	Avoids gambling rules and harm
Real-time clock and opening hours	The same, matched to Ghana time	Same feel, local rhythm
Taxes tracked weekly, wage policies, degree bonuses	Tax, degree bonuses and the NSS allowance as real mechanics	School and service matter
Other cities (Abuja, Port Harcourt) announced	All 16 regions on one map	One engine, no redraw
Guest browsing, an account to enter venues, 18+	The same, plus voice chat and marriage between players	More ways to connect
A reset after a fake-money exploit	Server-side money checks from day one	Prevents the problem

Planes, specifically. Players book flights from the phone, check in at the airport scene (skippable), and watch a cinematic flight over the real map. Flights connect Accra with other regions. Players can work as cabin crew, pilots or ground staff, and friends on the same flight share a voice room. Airlines sponsor flights. Why: it turns travel into an experience, and gives aviation careers a purpose.

Sources: Lagos Life guide: map, jobs, money, houses and rent, governor election, what is Lagos Life

Life systems

Time. The game day follows the real clock, so mornings and evenings match Ghana. Why: a shared clock makes the world feel alive and gives players reasons to log in together.

Needs and stats. Launch with Energy, Food, Hygiene, Mood and Vibe (community standing). Add Social, Health, Skill and Peace next, and ageing and family later. Why: too many bars confuse new players, and each new one needs content to feed it. Money is Sika, in cedis.

Starting paths. Nothing is compulsory. Players can begin as a senior high student, a university student, an apprentice, or go straight to work or business. Each career shows a requirements card before the player commits, with the steps, time, cost and success chance. Why: it teaches the real steps and keeps every choice meaningful.

Career	Steps	If there is no job
Doctor	Senior high school, university medicine, exams, national service, licence, then hospital or clinic	Locum shifts, teaching, side hustle
Lawyer	Senior high school, university law, law school, bar call, then a firm or own chambers	Paralegal work, writing, small business
Engineer	School, university or technical school, internship, then a firm or workshop	Repair jobs, freelancing, trade work
Teacher	Training college or university, national service, then a school	Private tutoring
Tailor, mechanic, hairdresser	Find a master, learn over time, open a workshop	Work from home, rent a chair, sell online
Trader or farmer	Capital, a stall or land, supplier contacts	Start smaller, partner with a friend
Creator or musician	Practise, post, build an audience, perform	Sponsored content, event gigs

Steps are compressed so a path takes weeks, not years.

Student life. Hall or hostel life, lectures, solo or voice study groups, quizzes, fees, scholarships and part-time jobs. Results come from attendance, study time and quiz scores.

National service (NSS). Graduates are posted to a sector and region, with a monthly allowance, supervisor sign-off, and sometimes late payments that push players into side hustles. The allowance follows the real figure: the latest approved amount I found is GH₵715.57 a month (from January 2023), and a raise to GH₵1,200 was proposed in 2026 but reported as not yet in effect. Make the amount a setting and check the current figure before launch. Keep the portrayal gentle and funny, and never accuse real offices or people. After service players can be retained, job hunt, return to school or start a business.

The hustle layer. If a job does not come, players hustle through odd jobs, selling, tutoring or deliveries, so nobody is stuck.

Story mode. An optional guided path of short chapters in the shared world: 1. Arrival (pick a start, find a room, get a first job, pay first rent). 2. Hustle (earn Sika, make three friends). 3. Community (join a circle, attend an event, mentor someone). 4. Love and home (court, marry, set up a home). 5. Power (run for office or start a business). Open mode has no goals except your own. Why: story mode helps newcomers, and open mode keeps the freedom.

Career circles. When a player picks a field, coworkers, bosses, events and after-work spots match it, and experienced players can mentor others. Each field shows short, checked "did you know" tips, labelled as general information, not professional advice. Why: it makes the world believable and gives players real-life value.

Getting around: transport and booking

How Lagos Life handles it, from public information only: players ride danfo buses across a map of real Lagos landmarks, and reports say they can also use taxis, motorbikes and tricycles, or drive themselves. The city has 28 named places in 8 areas, split into mainland and island, and each place has opening hours, prices and activities. A job in another area means travelling there. I could not find how the map is rendered or the exact booking steps, so the design below is ours, not a copy.

Mode	How it works in Ghana Life	Phase
Walk	Tap a spot and your avatar walks the real road. Free, slow and tires you	0
Ride app	Book from your phone, see the fare and wait, a driver arrives, you ride, then pay and rate. Most expensive, door to door	1
Taxi	Hail one on the street and haggle over the fare in a quick mini game	1
Trotro	Go to a station, wait until it fills, ride a fixed route, and pay the mate. Cheapest and slowest, with transfers at hubs like Circle	2
Tricycle and motorbike taxi	Short, cheap hops, with a chance of delays and fines	2
Bicycle or own car	Buy one, pay for fuel and repairs, and park it	3

Mode	How it works in Ghana Life	Phase
Intercity bus and domestic flight	Between regions, costing more Sika and time	4

Booking flow for a ride

1. Open the phone and choose Rides.
2. Pick a destination on the map.
3. See the fare and the wait, which depend on distance, time of day, rain and traffic.
4. A driver accepts, and you watch the car come along real roads.
5. Ride, or skip ahead to save time.
6. Pay Sika, then rate the driver.

Player-run transport. Players can work as taxi drivers, ride drivers, trotro drivers or mates, and earn fares from other players. When no player is free, a non-player driver steps in so nobody waits forever. Why: it makes transport a real job and a social meeting place, which Lagos Life's public descriptions do not mention.

Feel. Ride times are compressed to seconds or a minute, with a skip button. Rush hour, rain, power cuts and fuel price events change speed and cost. Why: it gives the city rhythm without wasting players' time.

Routing. Roads come from the same OpenStreetMap data as the map. Start with a precomputed road graph for Accra and simple pathfinding in the browser, and add a routing service later if needed. Why: it needs no paid service and keeps cars on real streets.

Anti-abuse. Fares are paid in Sika only, with caps, ratings and report buttons for harassment. Why: it keeps the driver and rider roles safe and fair.

Flights and cinematic travel

Lagos Life's public guide lists its airport as "coming soon", so I could not verify a plane scene there or how it was built. The design below is our own.

The idea. A flight is a short cinematic. The camera follows a plane over the real map from takeoff at the airport to landing in another region, so players see Ghana from above, passing over cities, roads, rivers and coast. Why: it makes travel feel like an experience and shows off the real map.

How it works

1. **Book.** Players book from the phone: pick a destination, see the fare, and check in. The airport scene can be skipped. Why: it matches the ride flow players already know.
2. **Camera.** The map camera tilts and turns behind the plane, using the map library's camera movement (pitch, bearing and animated moves). Why: it gives the sense of flying with no extra engine.
3. **Path and pace.** The plane follows a smooth line between airports, with a takeoff, a cruise and a landing, shortened to about 30 to 60 seconds with a skip button. Why: it feels real without wasting time.

4. **Look.** Raised buildings near the airport, colourful land, clouds drifting at different speeds, the sun matched to the game clock, city lights at night, dusty haze in the dry season and rain clouds in the wet season. Why: the sky changes with the world, so every flight looks different.
5. **The plane.** A small 3D model shown through a 3D layer, or a simple sprite in light mode. Use a generic fictional airline look unless an airline agrees to be shown, and check every model's licence. Why: it avoids trademark problems and keeps downloads small.
6. **Sound.** Engine sounds and cabin announcements in English and local languages, recorded by our own voices. Why: it adds a strong sense of place.
7. **Together.** Friends booked on the same flight share a cabin voice room. Why: a flight becomes a social moment.
8. **Light mode.** Slow connections skip the cinematic and show a simple progress bar. Why: it protects players on limited data.

Money. Airline sponsorships (fictional or agreed in writing), premium seat cosmetics and branded cabin screens. Why: it adds income without selling an advantage.

Cost and risk. Flying over satellite imagery loads many map tiles and uses a lot of data, so the stylised map is the default. Test on low-end phones before launch. This is a Phase 4 feature, built after the basics work.

Social systems

Live chat. Players near each other see chat bubbles in real time, with private chat, friend requests, online status, gifts of Sika and event invites. Chat accepts English, Pidgin, Twi, Ga, Ewe and more. Report, block and mute work everywhere, with a word filter and rate limits from day one.

Nearby voice. Players hear others within a short distance on the map, louder when close. Mic is off by default, with push to talk. Private voice rooms work for couples, crews and study groups. Voice launches after text chat is stable. Why: it is the standout feature, but it is the hardest to moderate and the last to build. Revision 2.1: private voice rooms for friends, couples and study groups launch earlier, in Phase 1 (task T13a), with push to talk, a small room cap, a kill switch and the 18+ rule. Nearby voice on the open map stays in Phase 2.

Marriage between two real players. Both players must agree at every step, and either can leave safely at any time.

1. **Courtship.** Dates at venues build a shared Affection meter.
2. **The knocking.** One player sends a formal request, and the other accepts, delays or declines, with optional witnesses.
3. **The list.** Both receive a generated list of items to fund, scaled to status, which they can split.
4. **The wedding.** A scheduled event where friends attend and give gifts.
5. **The household.** Pooled or separate wallets, a shared home, and a Peace of Mind boost when both stay active.
6. **Family events.** In-law visits, joint investments and naming ceremonies, decided together.

Safeguards: both players are verified 18 or older, divorce uses a cooling-off period and a clear rule for splitting shared Sika, and blocking or reporting a spouse always works.

Elections and the presidency. The ladder runs from neighbourhood leader to market association head to district assembly member to president. Players found and run fictional parties, with campaign seasons, rallies, live-chat debates and real player votes. Winners can set limited in-game rules and fund districts. Term limits and a recall vote prevent permanent rule, and money cannot buy votes. The people in the presidency building are fictional, so no real leader appears.

Open lives. Players choose any path: entrepreneur, politician, creative, street crew, bad boy or bad girl, community leader. Crime is a game mechanic, not a how-to.

- Crews, parties and rivalries use fictional names, never real gangs, tribes or religions.
- No real instructions for drugs, weapons, fraud, hacking or violence.
- Fights are consensual and limited to a clear mini game.
- Consequences are real in the game: police, fines, jail, lost Vibe.
- The game is 18+, with a clear content label.

Anonymity and the Reveal Ladder

The idea. Everyone is anonymous by default: a nickname and an avatar, no real name, no photo. Players from Kasoa to Kumasi to Tamale can meet, travel and become friends in the game, and nobody knows who is who until a person chooses to reveal more. Why: it feels safe and free, it is very Ghanaian to meet people across towns, and we store almost no private data.

Day-name identity. A new nickname starts from the Akan day-name (Kofi, Ama, Kwame, Afua and so on, with Ewe and Ga versions added) plus a random word, such as "Kofi of the Lagoon". Check the names with Ghanaian speakers before using them.

The Reveal Ladder

Rung	What is shared	Rules
1	A first name or a nickname	Both people agree. Either can step back at any time
2	The home town or region (for example Kasoa or Kumasi), never an exact place	Both agree, after a short wait
3	Voice: a private call	Both agree. The phone shows only nicknames
4	A social handle, typed in chat by the player	The game does not host or store photos in early versions
5	Meeting in real life	Not built until the plan says so. Groups only, public places, daytime (see the risk register)

- Every rung needs both people to say yes, and each has a short cooling-off wait (for example 10 minutes before rung 2 and a day before rung 4). Why: it stops rushing and slows scammers.
- Reveals can be taken back. A person can stop sharing, block or report at any time.

- The game never asks for real names, phone numbers, mobile money details, ID or photos. Warnings pop up when someone types words like "send money" or a number.
- Anonymous but accountable: the server knows the account so it can ban abusers, but not the person's real identity.
- Trust levels: brand-new accounts cannot start private chats with strangers until they have played for a while.
- 18+ only, with a clear safety page, and a lawyer review of age checks and privacy (including Ghana's data protection law) before public launch.

Ghana culture calendar

The idea. Real Ghanaian dates become events in the game: a festival scene on the map, matching outfits and music, small gigs, a learning note about the tradition, and a reason for friends to log in together. Dates live in a simple config file, so adding or changing one never needs new code.

Event	Where or who	Typical timing (verify each year)	In-game idea
Independence Day	Nationwide, Black Star Square in Accra	6 March	A parade on the map, flag outfits, night fireworks, fictional speeches
Eid festivals	Muslim communities nationwide	Moves with the moon	Greetings, festive outfits, feast scenes, gifting
Easter and Christmas	Nationwide	Easter moves, Christmas on 25 December	Church and family scenes, beach parties at Labadi, lights
Aboakyir	Winneba (Effutu people)	Usually early May	A festival town with drums and a durbar. No hunting game
Homowo	Ga people, Accra	Usually around August to September, announced yearly	A street feast scene, with food and respectful learning notes
Chale Wote	Jamestown, Accra	Usually August	Street art, music and dance, with art slots for creators
Panafest and Emancipation Day	Cape Coast and Elmina	Usually July to August	A heritage trail with history quizzes and performances
Fetu Afahye	Cape Coast	Usually early September	A colourful street procession and a durbar scene
Hogbetsotso	Anlo-Ewe, Volta Region	Usually early November	Drumming and dance, with Ewe phrases

Event	Where or who	Typical timing (verify each year)	In-game idea
Damba	Northern Region	Moves with the Islamic calendar	Drumming, northern smock outfits, northern phrases
Farmers' Day	Nationwide	First Friday of December	Market and farm gigs, prizes for farmers
Akwasidae	Asante	About every 42 days	Learn-only notes, with no royal rites as gameplay

Respect rules. Sacred rites are never mini games, jokes or sold items. Each festival scene is reviewed by people from that community before release, and credits the source. If a community objects, we change it. Dates above are typical, not guaranteed. Check each one with official or local sources every year.

The in-game phone

The idea. A phone on screen holds most of the social and money features. It is built from plain page elements on top of the map, not drawn in 3D, so it costs almost nothing on any tier.

App	What it does	Phase
Settings and Safety	Quality tier, data saver, report, block, mute, and Reveal Ladder controls	0
Contacts and Chat	Friends, private chat, requests	1
Calls	Private voice calls and small rooms (task T13a), showing nicknames only	1
Wallet	Sika, rent, gifts and susu circles	1
Rides	Book a ride (flow in "Getting around")	1
Jobs	Job board and gigs	1
Events	The culture calendar, RSVPs and reminders	1
Map and Passport	Routes, plus a Ghana Passport with stamps for regions, landmarks and festivals	1 and 2
Camera and Postcards	One tap makes an image and a link to share on WhatsApp	2
Radio and Music	Stations and artist slots	2
Mentors and Pen pals	Anonymous career questions and postcards between regions	2
Shop Codes	Discount codes from shops that claim their place	2
Brand apps	Sponsored apps from telecoms, food and fashion	3

Ghana magic: extra ideas, ranked

Each idea is mostly screens and server rules, so none should add lag. Impact and difficulty are my judgement.

Idea	Impact	Difficulty	Phase
Day-name identity (anonymous names)	High	Easy	0 to 1
Agoo and Amee: knock to enter a room, answered with "Amee"	Medium	Easy	1
Reveal Ladder	High	Medium	1
Private voice rooms ("voice parties")	High	Medium	1
Culture calendar events	High	Medium	1
WhatsApp postcards	High	Easy	2
Ghana Passport stamps	Medium	Easy	2
Susu circles (friends pool and rotate Sika)	Medium	Medium	2
Shop codes for real discounts	High	Medium	2
Anonymous mentors and pen pals	Medium	Medium	2
Twi, Ga and Ewe phrase learning	Medium	Medium	2
Map-a-thon: players fix missing shop names and hours	Medium	Medium	2
Diaspora "returnee" start	Medium	Easy	2
Dumsor blackout wave as a shared event	Medium	Medium	2

Scope guard: the rule of three

There are now many ideas, and a free game that tries to do everything will lag, confuse players and never finish. So the public launch leads with only three signature features, and everything else waits.

- Signature 1: the alive map (traffic, planes, day and night, quality tiers).
- Signature 2: anonymous friends with the Reveal Ladder and private voice rooms.
- Signature 3: the Ghana culture calendar with its first three events.
- Later: marriage, elections, businesses, land, real flights, regions beyond Accra, crime play and the creator marketplace.
- Rule: a new idea only goes in when another one comes out or moves to a later phase. Each phase adds at most three new signature features.
- Rule: any feature that makes Light worse than its budget waits until it is fixed.

Making money

The game is free for players. Income comes from businesses, artists and optional extras, as it did for Lagos Life, which reported about 46,900 US dollars in its first four days, mostly from advertisers. We copy the idea behind each of its earners, price it lower, and add more. Prices below are written in dollar equivalents. Set them in cedis at the current exchange rate, which I could not verify today.

Income source	Lagos Life (reported)	Ours	Why
Roadside billboard, per week	About 3.77 dollars	About 2.50 dollars	Cheap for small sellers
Large plot or board, per week	About 13.56 dollars	About 10 dollars	Mid-size businesses
Music in clubs	About 37.74 dollars	About 28 dollars	Artists want new listeners
Brand app on the in-game phone	A delivery app appears in the phone	Telecoms, food delivery, fashion and others, priced by quote	Players use the phone all day
Premium main board or event sponsor, per week	Not detailed	Up to about 35 dollars	One sponsor can cover an event

Reports differ on Lagos Life's lowest billboard price (about 5,000 naira per week in one report, from about 50,000 naira in another), so check their site before relying on these comparisons. Start our prices low while the audience is small, then raise them as players grow, and show buyers a report of views.

Claim your place. Every real place on the map starts with a generic name. Owners claim their real spot, add their name, logo, menu and banners, and pay for premium features. Why: it gives businesses a reason to join, protects us from implying a business has joined when it has not, and gives us income.

Self-serve banners. A business uploads a banner, picks a district, tier and dates, and pays by mobile money. Every banner is reviewed against the house rules before it goes live. Food sellers get a "Today's special" slot with a WhatsApp order link. There is a cap per street so the world stays beautiful.

Extra income ideas

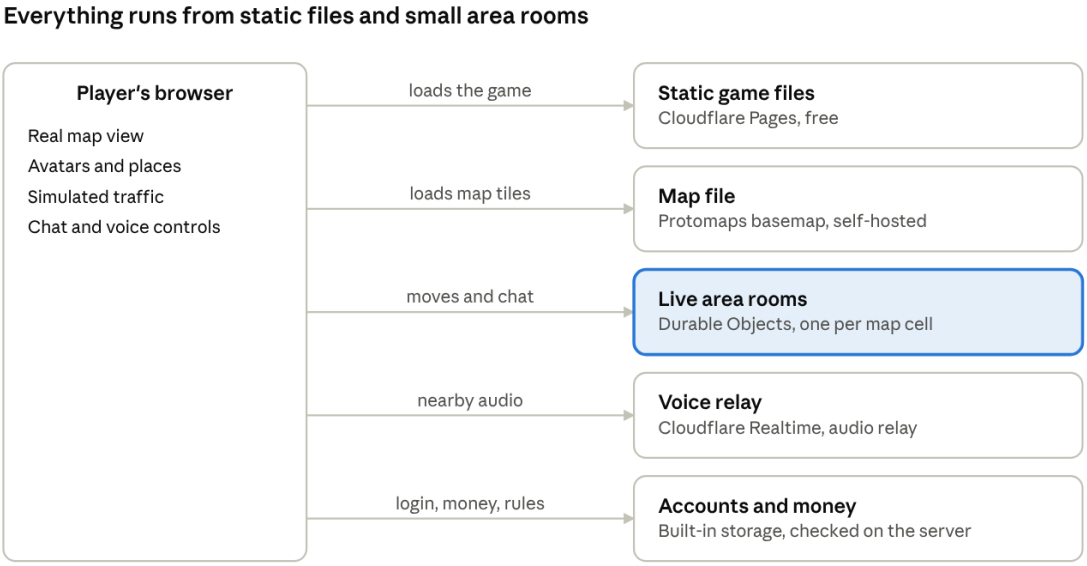
1. Game centres and arcades with sponsored tournaments.
2. Club and event tickets, with a share for us.
3. A jobs board where employers pay to post real vacancies.
4. Property listings from agents and developers.
5. School and university promotion.
6. Radio and playlists for artists and stations.
7. Cosmetics such as kente outfits, house styles and emotes.
8. A creator marketplace where artists sell designs and we keep a share.
9. Premium voice rooms: free small rooms, paid private and larger rooms.

10. Tips and a "support the game" button.

Rules. Never sell Sika for cash out. Never let money buy election results or marriage outcomes. Keep ads clearly labelled. Betting, alcohol, loans and health claims have special rules, so check them before accepting those ads. Only sell what you can deliver. Why: it protects players and keeps the game's reputation.

Tech and capacity

The game is a web app. The browser draws the map and runs the traffic simulation, and small server rooms handle only what must be shared: positions, chat and money.



architecture · browser plus 5 services

The browser loads the game and the map as static files, and talks to a live room for the area it is in. Voice and accounts go through their own services. All money and rules are decided on the server, never in the browser.

Part	Choice	Why
Map view	MapLibre GL JS with a Protomaps basemap of OpenStreetMap data	Free, open source, self-hostable
Game logic in the browser	TypeScript	Common, and Claude writes it well
Game files	Cloudflare Pages	Unlimited bandwidth for static files

Part	Choice	Why
Live rooms	Cloudflare Workers with Durable Objects	Built for many connections in small groups
Storage	Durable Object storage first, a database later	No pause risk, and can grow
Voice	Cloudflare Realtime (SFU and TURN)	1,000 GB of audio free each month, then 0.05 dollars per GB
Payments	A mobile money provider that supports Ghana business accounts, to be chosen	Players and sponsors pay in cedis

Capacity plan

1. **Area rooms.** The map is split into cells, and each cell is one live room. A player joins their cell and sees its neighbours. Why: the server only handles the people who matter to each other.
2. **Room cap and copies.** Cap a room at a set size (start with about 100, then measure), and open extra copies of busy places. Why: no single room gets overloaded.
3. **Waiting line.** When the city is full, show a friendly queue instead of letting the server fail. Why: players already inside keep a smooth game.
4. **Send less.** Share positions only with nearby players and a few times per second. Why: fewer messages means lower cost and less lag.
5. **Shared traffic seed.** Traffic comes from a clock and seed, so it costs no network data. Why: it looks busy at no cost.
6. **Server checks.** Money, jobs, rent and marriage are decided on the server, with rate limits. Why: it blocks cheating, and Lagos Life's fake-money exploit shows what happens without it.
7. **Test and alert.** Test with friends before inviting the public, and set usage alerts. Why: free limits fail suddenly, so you want a warning first.

Scale plan: from 100 players to 1 million

Honest framing. "1 million users" can mean three different things. One million registered accounts is a realistic design target. One million players online at the same moment is a different world and is not a goal. For a game like this, the number online at the same time at the busiest hour is usually a small slice of the registered users (my guess is 1 to 5 percent, so 10,000 to 50,000 at the peak for 1 million accounts; measure it). Nobody can promise the game never crashes. What we can do is design it so that growth is controlled and overload turns into a waiting line, not a crash. It is free while small and cheap in proportion when large, but it is not free at 1 million.

The big alternative: static-first, client-heavy, many small rooms

- Static-first. The game, the map and the content are static files on Cloudflare Pages, which has unlimited bandwidth for static files. Downloads, the largest load from a million people, cost nothing.

- The player's device does the work. Traffic, planes, weather, crowds and day and night run on each phone from a shared clock and seed, so every extra player adds no server cost for these.
- Many small rooms, never one big one. Each map cell is a room of about 100 players, with copies ("channels") of busy places. A matchmaker puts each player into the least-full copy.
- One tiny server object per player for their own data (wallet, needs, job, friends). A million small objects cost almost nothing while nobody is playing.
- Send little. Click-to-move, nearby-only updates, compressed messages, batched writes, and idle rooms put to sleep.
- Control who gets in (admission control, below), so the game is never asked to carry more than it was tested for.
- Keep the server logic in plain TypeScript so it can move to another host if prices or terms change.

Rough cost model (estimate from Cloudflare's published rates, to be load tested)

Cloudflare's published rates say a live room costs about 0.6 US cents per hour while active (128 MB at 12.50 dollars per million GB-seconds), and WebSocket messages are billed at 1 request per 20 messages. In my rough sums, duration is far bigger than request cost. Free plan: about 100,000 requests and 13,000 GB-seconds a day. Paid plan: 5 dollars a month, with 1 million requests and 400,000 GB-seconds included. The table assumes each room is busy about 8 hours a day, 10 incoming messages per player per minute, and rooms 30 to 100 players full. It leaves out voice, storage, and other services.

Stage	Players online at the peak	Rooms	Rough monthly cost
Private test	Up to about 100	1 to 3	0 dollars (free plan)
Small launch	About 500 to 1,000	5 to 34	About 5 to 50 dollars
Growing	About 10,000	100 to 334	About 150 to 450 dollars
Big	About 100,000	1,000 to 3,334	About 1,500 to 4,600 dollars
One million online at once	1,000,000	10,000 to 33,334	About 15,000 to 46,000 dollars, not a goal

Why this is fine. A game with that many players should earn from banners, sponsors, claimed places and cosmetics (see "Making money"). Lagos Life reported about 46,900 US dollars in its first four days, though that is exceptional and cannot be promised. The aim is for income to arrive before the bills do, and for a spending limit to protect you if it does not.

Admission control: how the game never crashes

- Invite waves. Open to 100, then 500, then 2,000, then 10,000 and so on, each only after the last wave passes its tests.
- Waiting line with an honest estimate, and the "the city is resting" screen when the day's limit is used.
- Budget breaker. Our own code counts usage and estimated cost, and stops admitting new players when the monthly limit you set is reached. Do not rely on the host to stop spending for you.

- Kill switches for voice, notes, banners and heavy effects, so one feature cannot take the whole game down.
- Browse-only mode when overloaded: guests can still look around while logged-in actions wait.
- A status banner on the website, and alerts at 70 and 90 percent of any limit.

Realtime back ends considered

Option	Free limit	Scales to	Verdict
Cloudflare Workers with Durable Objects	About 100,000 requests a day free, paid plan from 5 dollars a month	Millions of players, by adding more small rooms	Primary choice
Colyseus (open-source game server)	The software is free, the hosting is not	As far as the server you pay for	Fallback: the same TypeScript logic can move to it
Firebase Realtime Database	100 simultaneous connections on the free plan	Large, but costly	Too small for free
Supabase Realtime	200 connections, and free projects pause after 7 days	Medium on paid plans	A possible later database, not for live rooms
Hosted realtime services (Ably, Pusher, PubNub)	Small free tiers, not checked in detail	Large, billed per message and connection	Backup idea, check prices first
A server at home	Free but must stay on	Very small	No

Features that scale well (they cost almost nothing per extra player)

Feature	Why it scales	Phase
Synchronised spectacles: midnight fireworks, festival parades, a New Year countdown	Runs from the shared clock on each device, so a million players can see the same show at no server cost	1
Ghost crowds	Simulated pedestrians from the shared seed, so quiet streets still feel busy and early rooms never feel dead	1
Town channels (Kasoa, Kumasi, Tamale and so on)	Players meet people from their own region first, in small rooms	1
Daily treasure of the day	One small request per player per day	1
Event seasons	Content from a config file, delivered by the CDN	1
Notes on the map	A player leaves a short, moderated note at a place that others read later, with no live connection	2

Feature	Why it scales	Phase
WhatsApp postcards	The image is made on the player's device and shared through WhatsApp	2
Leaderboards and festival scoreboards	Updated in batches every few minutes	2

Load testing. Write a bot script that opens fake players and ramps from 100 to 1,000 to 5,000, and record the frame rate on the test phone, server errors, messages per second and the cost. Run it on a test copy, never the live game. A large test will use up the free daily limit, so budget about 5 dollars for one paid month for the test.

Rules for Claude. Every new feature must state its cost per player-hour and whether it needs a live connection. No feature may need an always-on server for each player. Keep the server logic portable.

Risk register

This is a stress test of the plan, written as a sceptical engineer would. Likelihood and impact are my judgement (Low, Medium, High), not measured. Each risk lists the cheapest fix and the test or task that checks it. If a test fails, stop and report it before building more.

#	Risk	Likelihood / impact	Cheapest fix	Checked by
R1	Weak phones cannot draw the map, cars and avatars smoothly	High / High	Quality tiers, the performance budget, drawing only what is visible, and testing on a cheap Android	Test 1
R2	A map file is over 25 MiB, which is the Cloudflare Pages limit per file, or the host does not allow reading parts of the file	High / High	Cut the map into several smaller files, or find another free host that needs no card	Test 0
R3	The live server is far from Accra, so actions feel slow	Medium / Medium	Click-to-move and smooth animation hide delay. Check whether Cloudflare offers a location hint for Africa. Avoid fast-reaction gameplay	Test 3
R4	The free daily limit runs out during a rush of players	Medium / High	Usage counter, "city is resting" screen, waiting line, and the 5 dollar paid plan ready	T13, T21

#	Risk	Likelihood / impact	Cheapest fix	Checked by
R5	Links opened inside WhatsApp, Facebook or Instagram browsers run badly or fail	Medium / High	Detect the in-app browser and show an "open in Chrome or Safari" button	Week-one day 7
R6	Map data gaps in parts of Ghana: missing buildings, names or opening hours	High / Medium	Standard building height, generic names, our own opening-hours content, spot-checks per district	T9
R7	Fake money or cheating	Medium / High	All money and rules decided on the server, with rate limits	T21
R8	Chat abuse that one person cannot review	High / High	Word list, report, block and mute, text-only launch, a small invite-only audience first, trusted volunteer moderators	T5, T21
R9	A typed date of birth cannot prove 18+, and data protection rules apply	Medium / High	Clear 18+ label and rules, collect little data, lawyer review before any public launch	Verify list
R10	Players lose passcodes, with no email to recover them	Medium / Medium	Show a one-time recovery code when the account is made	T8
R11	A beginner builder is overloaded by scope	High / High	One task at a time, the week-one cut, save to Git after every task, keep this document updated	Every task
R12	Claude's free-plan message limits slow the build	Medium / Medium	Small tasks, short prompts, paste only the needed files, keep notes in the project folder. A paid plan is optional later	Ongoing
R13	Voice fails on some mobile networks	Medium / Medium	Use Cloudflare's relay service, test early on MTN, Telecel and AT, keep voice optional	T14
R14	Data cost puts players off	Medium / Medium	Light mode, caching, a data-saver switch, showing approximate data used	Budget
R15	AI-made characters look uneven, with weak hands, faces or hairstyles	High / Medium	Compare several tools, pick the best, clean up in Blender, and test Ghanaian hairstyles early	Character test

#	Risk	Likelihood / impact	Cheapest fix	Checked by
R16	A free AI tool does not allow commercial use, or changes its free terms	Medium / High	Read each licence before use, keep a licence note with every asset, switch to hand-made or paid models if needed	Before launch
R17	Hair and clothes do not fit the shared skeleton, or poke through the body	High / Medium	Make parts fitted to the same body, hide covered body parts, keep the first outfit set small	Character test
R18	Real 3D characters make weak phones lag	High / High	Real 3D only on Ultra for the nearest players and in close-up scenes, sprites elsewhere	Test 1b
R19	Banned players come straight back, because accounts are free and anonymous	High / Medium	Trust levels for new accounts, rate limits, bans tied to the account and device clues, and later an optional email or phone check	T8, T21
R20	Anonymous chat enables grooming, scams or unsafe real-life meetings	Medium / High	Reveal Ladder waits, scam warnings, report and block, 18+ only, no real-life meetings in early phases, and meetings only in groups at public places later	Reveal task
R21	A festival scene offends a community or shows sacred things wrongly	Medium / High	Learn-only for sacred events, review by people from each community, easy to change or remove a scene	Calendar tasks
R22	Too many ideas, so nothing is finished	High / High	The rule of three, one task at a time, and a phase gate before anything new	Every phase
R23	Costs grow faster than income	Medium / High	Budget breaker with a monthly limit, invite waves, early banner and sponsor sales, and the waiting line	Scale plan
R24	Cloudflare changes its free limits or prices	Medium / High	Keep server logic in plain TypeScript so it can move (for example to Colyseus), and keep a fallback ready	Scale plan

#	Risk	Likelihood / impact	Cheapest fix	Checked by
R25	A viral spike overloads one popular place such as central Accra	High / Medium	Channels (copies of a room), town channels, the waiting line and ghost crowds in quiet areas	Load test

Costs

This section is for later growth only. The zero-dollar plan above costs nothing, and Claude must not add anything listed here (satellite imagery, paid plans, a managed database) to the build. Nothing stays free and crash-proof as thousands of players arrive. Free plans have hard caps, and when you pass one the service stops working. Figures come from search results dated July to September 2026, and I did not open every provider page, so verify each before relying on it.

Part	Free option	Free limit	When you outgrow it
Game files	Cloudflare Pages	Unlimited bandwidth for static files, 500 builds per month	Stays free for this part
Live rooms	Cloudflare Workers with Durable Objects, free plan	About 3 million requests per month, daily reset, and calls fail once used up	About 5 dollars per month for Workers Paid, then small pay-as-you-go charges
Voice	Cloudflare Realtime	First 1,000 GB per month	0.05 dollars per GB
Stylised map	Self-hosted Protomaps file	No licence fee	Storage and bandwidth only, check the host's free allowance
Satellite look (optional)	Mapbox	50,000 map loads per month, card required	About 5 dollars per 1,000 loads above that, falling at higher volumes
Database (optional)	Supabase free plan	200 live connections, projects pause after 7 days of no use	25 dollars per month for Pro
Art and sound	Your own work, or assets licensed for free commercial use	Check each licence	Artist fees if you hire

Part	Free option	Free limit	When you outgrow it
Legal	Writing your own rules	None	Lawyer and trademark fees, worth doing before marketing

Stage costs (estimates from published rates)

Stage	Players	Approximate monthly cost
Private test with friends	Up to about 100 online	0 dollars
Small public launch	About 200 to 500 online	About 5 to 30 dollars
Add voice for 1,000 daily players, an hour each	About 30,000 listening hours	Roughly 15 to 40 dollars extra, assuming about 32 kilobits per second of audio and three people audible at once
Satellite look for 5,000 players opening the game 30 times a month	About 150,000 map loads	Roughly 450 dollars, so switch it on only when income covers it
Viral spike	Many thousands	Cannot be estimated reliably, depends on design and income

Optional later spending, not part of the zero-dollar plan

Item	Cost	Why
Build and host on free tiers	0 dollars	Covers a test of about 100 players
Cloudflare paid plan, once the free limit is near	About 5 dollars per month	Stops the game failing as players grow
Own domain name (optional)	Roughly 10 to 15 dollars per year, check a registrar	The free address works, so this can wait
Reserve	About 30 dollars	Lets you react to a surprise

Satellite imagery needs a card on file and can grow quickly in cost, so it is not part of the zero-dollar plan. Claude's free plan has message limits, so building may go slower, and a paid plan is not required to start.

Cheaper later. Move live rooms to a self-hosted server only when income allows, keep audio nearby-only, keep light mode on by default for slow data, and send updates only to nearby players. Why: each of these cuts the main costs, which are live connections, audio and map loads.

Scale costs. For what each stage costs from 100 players to 1 million, see "Scale plan: from 100 players to 1 million".

Legal, safety and credit

I am not a lawyer. Treat this as a working checklist and have a Ghanaian or Nigerian IP lawyer review it before launch.

Staying original. Copyright protects specific expression such as art, text, code and audio, not the idea of a life game. Trademarks protect names and logos.

Safe to do	Avoid
Build a multiplayer life game with needs, jobs, rent and a map	Copying Lagos Life's code, art, layout, icons, text or jokes
Use real Ghana places, food and customs as inspiration	Using its names, such as "Nepo" or "Lapo", or a similar game name and logo
Use common words like Vibe, Sika and hustle	Implying we are official, affiliated or an "unofficial sequel"
Use original music and sound	Using real songs without a licence
Compare features honestly in marketing	Real celebrities, politicians or private brands as characters without permission

Credit. You do not need to credit Lagos Life in the game. A casual thank-you in interviews is a courtesy. Never imply a connection, and if you ever reuse anything specific of theirs, get permission first.

Data and imagery licences

Source	What to know
OpenStreetMap and Protomaps basemap	Credit OpenStreetMap on the map. Share-alike can apply to derived map databases, so keep map data separate from game data
Protomaps hosted service	Free for non-commercial use only, so self-host for a commercial game
Overture places data	Permissive licence, but mixing it with OpenStreetMap data can bring in the share-alike rule
Google satellite imagery	Do not use. Google restricts commercial use and tracing of roads from it
Mapbox satellite	Read its terms for use in a game before switching it on
OpenAerialMap	Check the licence on each image, because some are not CC-BY

House rules before play. Every player reads and accepts the rules before entering, and again when they change:

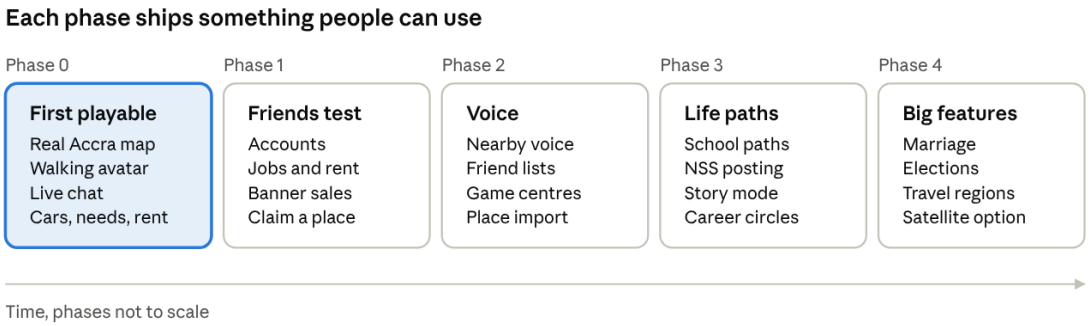
1. No hate speech, threats, harassment or sexual content aimed at others.
2. No sharing of real phone numbers, mobile money details, passwords or addresses.
3. No real scams or requests for real money. Sika has no cash value.
4. No attacks on real people, tribes, religions or parties.
5. Be 18 or older.
6. Crime and gang play stays fictional and consensual.

Word flags. A blocked and warning word list in Pidgin, Twi, Ga, Ewe and English, written by Ghanaian speakers, with spelling tricks handled. A first hit blocks the message and warns, repeats mute, and persistent abuse leads to suspension.

Safety, privacy and moderation

1. 18+ with a date of birth check on the server. Why: chat plus romance with minors is a serious legal and safety issue.
2. Report, block and mute on every player and message, with a human review queue. Why: automated filters miss abuse.
3. Tell players before voice that reports may use a short clip of recent audio, and check the privacy rules with a lawyer, including Ghana's data protection law.
4. Keep Sika non-withdrawable and never allow cash out. Why: it avoids gambling and finance rules.
5. Scam warnings in the game about real mobile money details.
6. Check advertising rules for betting, alcohol, loans and health claims, and payment and tax rules before taking real payments.

Roadmap and start-now checklist



roadmap · 5 phases

Each phase ends when its gate is met: Phase 0 when about ten friends can play at once without lag, Phase 1 when real businesses buy banners, Phase 2 when voice works for small groups without abuse, Phase 3 when new players finish a story chapter, and Phase 4 when each big feature has been tested by real players. Why: a gate keeps you from adding features on top of a broken base.

Start now

1. Make free accounts on GitHub and Cloudflare, and keep your Claude account.
2. Install Node.js and a code editor such as VS Code.
3. Ask Claude to guide you through the first prompt below, one step at a time, testing each step before the next.
4. Download a cut of the Protomaps basemap for Greater Accra with the free Protomaps tool, and host it yourself. Claude can give you the exact commands.
5. Deploy to Cloudflare Pages, share the link with five to ten friends, and watch what they do.
6. Write down every bug and idea, then fix the biggest one first.

First prompt to give Claude

```
I am building a free browser life game set in Ghana, with Accra as the default home.
I am a beginner, so explain every step in plain words and wait for me to confirm each
step works.
Step 1: Give me the folder structure and exact commands to set up a TypeScript
project with MapLibre GL JS.
Step 2: Show a real map of Accra using a self-hosted Protomaps PMTiles file, with my
own bright colour style.
Step 3: Add a player avatar that walks along the roads when I tap or click.
Step 4: Add live text chat between players in the same map area, using Cloudflare
Workers and Durable Objects.
Step 5: Add a few simulated cars that move along real roads from a shared clock and
seed.
Rules: keep all money logic on the server, credit OpenStreetMap on the map, and keep
everything free to run.
```

Documents to keep in the project folder: a short vision and rules file, tech notes, a content bible of places, jobs, events and slang in your own words, a task list, and a test checklist. Why: Claude can then pick up where it left off each time.

Prompting habits. Ask for one small feature at a time, paste exact error messages, ask Claude to explain code in plain words, and ask for security checks on money, chat and accounts before launch.

Week-one sprint

Honest scope. The whole plan cannot be built and polished in one week, even with Claude's help. One week can produce a tech test that proves the hard parts: a styled real map of Accra that is smooth on a cheap phone, with walking, live chat, moving cars, an ambient plane and quality tiers. If these work, the rest is adding features on a proven base. If one fails, we find out in days, not months.

Four go or no-go tests

Test	What to do	Pass if (targets)
Test 0: map hosting	Cut a small Accra map, put it on Cloudflare Pages and load it on a phone	Every file is under 25 MiB and the map loads

Test	What to do	Pass if (targets)
Test 1: cheap-phone speed	Styled Accra map plus 15 moving cars, on the cheapest Android you have, in Light	About 30 frames per second, and the map shows in about 5 seconds on mobile data
Test 1b: characters	Load 10, 20 and 30 animated low-poly 3D characters with the map on the cheapest Android	10 characters hold about 30 frames per second, or real 3D stays in close-up scenes only
Test 2: live movement	Two phones in one room, one taps to walk	The other phone sees the walk start in about a second, with no errors
Test 3: distance to server	Measure round-trip time from Accra to the live room on mobile data	Typically under about 250 milliseconds. If worse, redesign before building more

Day by day

Day	Tasks	At the end of the day you should see
1	T0 and T0a (Test 0)	A hello page live online, and the map-file result written down
2	T1 and T2	A styled Accra map on a computer and a phone, with the OpenStreetMap credit
3	T2a and T3	The Light, Standard and Ultra switch with a frame-rate overlay, and an avatar walking on real roads
4	T4 (Tests 2 and 3)	Two phones see each other walk, and the delay is measured
5	T5	Chat bubbles, a word filter and a report button
6	T7 and T7a (Test 1)	Moving cars and an ambient plane, with Light inside its budget
7	Friend test	Try with 5 to 10 friends aged 18 or over, including the WhatsApp browser. Fix the five biggest problems and update this document

If you fall behind. Drop things in this order: the plane (T7a), then cars above 15 in Light, then chat polish. Never drop the tests. The results of day 6 and 7 decide week two.

Not in week one. Needs, job and rent (T6), accounts and the age check (T8), places (T9), jobs, rides, businesses, voice, marriage, elections, real flights and regions beyond Accra. Why: each needs a proven base first. Share the week-one test only with adult friends until T8 adds the age check and house rules.

Prompt to paste for any task

Here is my Ghana Life master plan. Do task T__ from the task list in the Build brief. Follow the hard rules and the performance budget. Explain every step in plain words, give exact commands, and wait for me to confirm each step works. When finished, run the "done when" test, report the result, and tell me what to write in the risk register.

Prompt to stress-test the plan

Here is my Ghana Life master plan. Act as a sceptical senior game engineer. List the 10 biggest technical risks to a smooth experience on cheap Android phones on mobile data in Ghana. For each, give how likely it is, how bad it would be, and the cheapest fix. Do not repeat risks already in the risk register, add new ones.

Build brief for Claude

Your role and goal. You are building Ghana Life, a free browser life game set in Ghana with Accra as the default home, as described in this document. The person you are helping is a beginner, so explain every step in plain words, give exact commands, and wait for them to confirm each step works before the next. Work on one task at a time from the task list.

Hard rules

1. The first version must cost zero dollars and need no payment card. Do not add any service that needs one.
2. Stay inside the free limits in the zero-dollar plan, and follow its design rules.
3. Keep all money, jobs, rent and marriage logic on the server, never in the browser.
4. The game is 18+. Players must confirm age and accept the house rules before playing.
5. Use original names, text, art and code. Never copy Lagos Life's code, art, text or names.
6. Credit OpenStreetMap on the map. Keep map data separate from game data.
7. In-game money (Sika) has no cash value and cannot be withdrawn.
8. Crime and gang play is fictional and consensual, with no real instructions and no real groups.
9. Before each task ends, run its "done when" test and report the result.

- Follow the performance budget in "Quality tiers and performance". A task is not done until the Light tier stays inside its budget on a cheap Android phone.
- Load heavy things (3D, voice, flights) only when needed, and never in Light. Keep every file in the Pages site under 25 MiB.
- If a test in the week-one sprint fails, stop and report it. Do not work around it silently.
- Anonymity: store as little personal data as possible. Never ask for real names, photos, phone numbers or ID. Every reveal is optional, mutual and can be taken back.
- Do not build real-life meetups, marriage, elections or crime play until the plan says so. Follow the rule of three.
- Every new feature must state its cost per player-hour and whether it needs a live connection. No feature may need an always-on server per player. Keep server logic portable.

Tech stack

Layer	Tool	Notes
Language	TypeScript for browser and server	One language, easier for a beginner
Build tool	Vite	Fast and simple
Map	MapLibre GL JS with the PMTiles protocol	Free and self-hosted
Map data	Protomaps basemap cut for Greater Accra, made with the pmtiles command-line tool	Built from OpenStreetMap data
Roads	A road graph file made by a script from OpenStreetMap data, with A* pathfinding in the browser	No paid routing service
Live server	Cloudflare Workers and Durable Objects with built-in SQLite storage and the WebSocket hibernation API	Free plan
Hosting and deploy	Cloudflare Pages and the Wrangler command-line tool	Free, no card
Voice (Phase 2)	WebRTC with Cloudflare Realtime	First 1,000 GB a month free
Places	Overture Maps places and OpenStreetMap, cleaned by a script	Free open data
Tests	Vitest, multi-tab manual tests, and a small load-test script	Free
On the laptop	Node.js LTS, npm, Git, VS Code	Free
People and vehicles	PixiJS drawn above the map, one shared sprite sheet	Light 2D drawing of many small sprites
3D (Ultra only)	three.js, loaded only on Ultra	3D plane and special effects, never downloaded by Light
Simulation	Web Worker for traffic and plane paths	Keeps the screen smooth
Quality tiers	Device check plus a frame-rate test, with player override	Weak phones get Light, strong ones get Ultra

Repository layout

```
ghana-life/
  README.md              how to run and deploy
  docs/MASTER_PLAN.md    this document
  apps/web/              browser game: Vite, TypeScript, MapLibre
    src/map/  src/game/  src/ui/  src/net/
    public/tiles/accra.pmtiles
  apps/server/           Cloudflare Worker and Durable Objects
    src/rooms/  src/rules/  src/moderation/
  data/                  map and places data, with build scripts
  scripts/               build the road graph, import and clean places
  content/               jobs.json, places.json, events.json, words.json (our own
writing)
```

Core data (stored on the server)

Record	Main fields
Player	id, nickname, passcode hash, age confirmed, created time, Vibe
Wallet	player id, Sika balance
Needs	player id, energy, food, hygiene, mood, last updated
Job	player id, career, level, performance, last shift
Home	player id, home, rent due, arrears
Place	id, generic name, category, position, opening hours, claimed flag
Banner	id, tier, district, image file, start, end, status
Report	id, reporter, reported, reason, time, status

Live messages

From the browser	From the server
hello (nickname, token)	snapshot of the nearby area
move_to (destination)	player joined, left, or started walking (path and speed)
chat (text)	chat message
interact (place, action)	needs, wallet and job updates
start_shift, book_ride	world events such as rain, power cuts and traffic, and notices

Task list

Do these in order. Each task ends only when its "done when" test passes. Phases match the roadmap.

#	Task	Done when	Phase
T0	Set up free accounts (GitHub, Cloudflare) and install Node.js, Git and VS Code	A "hello" page is live on a Cloudflare Pages address	0
T0a	Pre-flight tests: cut a small Accra map, check each file is under 25 MiB and loads from Pages, and measure round-trip time from Accra to a tiny live room	Both results are written in the risk register, and the map file loads on a phone	0
T1	Create the project with Vite, TypeScript and MapLibre, in the repository layout above	A blank styled map opens in the browser and on a phone	0
T2	Cut a Protomaps map of Greater Accra, host it as a static file, style it in bright colours, and show the OpenStreetMap credit	The map loads in under 5 seconds on a phone, and light mode loads fewer details	0
T2a	Add the quality tier check (device clues plus a 5-second frame-rate test), the Light, Standard and Ultra switch, a frame-rate overlay and the performance budget	The page picks Light on the cheap Android and Ultra on a laptop, and the overlay shows the frame rate	0
T3	Build the road graph from OpenStreetMap data, and make an avatar that walks along	The avatar follows real roads to any tapped place	0

#	Task	Done when	Phase
	roads when the player taps a destination		
T3a	Add a placeholder person (a coloured figure) with a walk animation, then run Test 1b with 10, 20 and 30 animated 3D characters	Frame rates are recorded on the test phone, and the result decides which tiers use real 3D	0
T4	Build the live server: one Durable Object room per map cell, with hello, move_to and snapshot messages, and hibernation	Two browser tabs see each other walk, and idle rooms stop using compute	0
T4a	Add channels: copies of a busy room and a matchmaker that puts each player in the least-full copy	Two copies of one room open when the first is full, and the matchmaker fills the least-full one	1
T5	Add chat bubbles, a word filter, and report, block and mute	Messages appear for nearby players, banned words are blocked, and a report is saved	0
T6	Add the real-time clock, five needs, one job with a shift, and rent, all decided on the server	Needs fall over time, a shift pays Sika, and unpaid rent builds arrears	0
T7	Add simulated traffic from a shared clock and seed, with cars, taxis, trotros, tricycles and bicycles	Two tabs show the same traffic, and light mode shows fewer vehicles	0
T7a	Add ambient planes on approach to and from Kotoka, driven by the	Two tabs show the same plane in the same	0

#	Task	Done when	Phase
	shared clock and seed (sprite in Light and Standard, 3D model on Ultra)	place, and Light stays inside its budget	
T7b	Build the first 60 seconds: loading screen, camera swoop, quick character start, landing scene and the guide message	A new player reaches their first walk within 60 seconds, in Light and in Ultra	1
T7c	Add ghost crowds: simulated pedestrians from the shared seed, and synchronised spectacles from the shared clock (fireworks at set times)	Two tabs see the same fireworks at the same time, and quiet streets look busy within budget	1
T8	Add guest mode, nickname and passcode accounts, the age check and the house rules screen	A new player must accept the rules before playing, and accounts survive a restart	1
T9	Import and clean places from Overture and OpenStreetMap, add categories, opening hours and 20 landmarks with simple interiors	Tapping a place shows its name, hours and activities	1
T9a	Add the culture calendar: a config file of events with dates, and the first three events (Independence Day, Farmers' Day, Christmas) as simple scenes	Changing a date in the config file moves the event, and each scene shows a learning note	1

#	Task	Done when	Phase
T10	Add the job board, careers with ranks and promotion, gigs, and the Sika wallet	A player can take a job, get promoted, and do a gig	1
T10a	Add the phone shell with Settings and Safety, Contacts and Chat, Wallet, Jobs and Events	The phone opens over the map, every app works, and Light stays inside its budget	1
T11	Add walking, ride booking and taxis with fares, waits and a driver animation	A booked ride arrives along roads and charges the right fare	1
T12	Add admin-placed banners and the claim-your-place flow, controlled from a config file	A banner appears on the map, and an owner's claimed place shows its real name	1
T13	Add friends, private chat and a daily limit guard that shows the "city is resting" screen	Friends can chat privately, and the guard triggers at 90 percent of the daily limit	1
T13a	Add private voice rooms for 4 to 6 players, muted by default with push to talk, a room cap, a usage cap, a kill switch and the 18+ rule	Friends in a private room hear each other, and the kill switch stops all voice	1
T13b	Add day-name nicknames and the Reveal Ladder: mutual requests, cooling-off waits, take-back, report and scam warnings	Two players climb two rungs with mutual agreement, and a take-back works at once	1
T14	Add nearby voice with push to talk, mute by	Two players hear each other nearby, and the	2

#	Task	Done when	Phase
	default, a cap and a kill switch	kill switch stops all voice	
T15	Add game centres, trotro rides with a mate, and player drivers	A player can drive, and riders pay them	2
T16	Add school, university, apprenticeship and NSS paths with the real allowance, and story mode	A new player can finish a story chapter, and a graduate gets posted	3
T17	Add homes and rent advance, businesses, land and the daily clue	A player can buy a business, and rent works weekly or in advance	3
T18	Add marriage between two players, with consent, divorce and safeguards	Two players complete courtship, wedding and a household	4
T19	Add elections and leadership offices with fictional parties	A full election runs, and a recall works	4
T20	Add the airport, flights and the cinematic view, then more regions	A flight takes a player to another region with a skippable cinematic	4
T21	Run a safety and load review: security checks on money, chat and accounts, and a test with friends	No way to create free Sika is found, and ten players play at once without lag	1 onward
T21a	Add admission control: invite waves, waiting line, budget breaker and kill switches for voice, notes and banners	The breaker stops new players at the set limit, and each kill switch turns its feature off at once	1

#	Task	Done when	Phase
T21b	Write a bot load test that ramps from 100 to 1,000 to 5,000 fake players on a test copy, and record frame rate, errors, messages per second and cost	The results are written in this document and set the real room size and monthly cost	1

Definition of done for every task. The code runs, the "done when" test passes, the free limits are not exceeded, no new paid service or card is needed, and the change is saved in Git with a short note.

Verify before launch, open decisions and sources

Verify before launch

- ☐ Current NSS allowance (last approved GH₵715.57, with a GH₵1,200 proposal)
- ☐ Today's cedi exchange rate, then set banner prices in cedis
- ☐ Mapbox terms for satellite use in a game, if you add it
- ☐ Free limits and prices for Cloudflare, Mapbox and any database
- ☐ A mobile money provider that supports Ghana business accounts
- ☐ Trademark search for the game name and logo
- ☐ Lawyer review of licences, privacy, payments and advertising rules
- ☐ Spot-check the map and places in each district

Open decisions

- ☐ Game name and logo (Ghana Life is only a working title. It copies the pattern of "Lagos Life", so choose a clearly different name before launch)
- ☐ Who builds it: you, a freelancer, or a small team
- ☐ Start picked by the player or drawn by the server
- ☐ Voice: private rooms in Phase 1 (decided), nearby voice in Phase 2
- ☐ Whether to add a satellite look, and when income allows
- ☐ First region beyond Accra
- ☐ Final choice of 2D engine after the cheap-phone test (PixiJS, or drawing vehicles as map symbols)
- ☐ Where to host map files if a cut is over 25 MiB (several smaller files, or another free host)
- ☐ Which cheap Android phone is the official test phone
- ☐ Who finishes the base bodies: you in Blender, a freelance 3D artist, or ready-made licensed models
- ☐ How Light gets its character sprites: baked from the 3D models, or separate layered 2D art
- ☐ Which AI tool makes the first base bodies (compare Hunyuan 3D, Meshy, Tripo and Rodin)
- ☐ Whether real-life meetups ever exist, and under what rules (proposed: not before Phase 3, groups only)

- ☐ Who reviews each festival scene from the community, and how to credit them
- ☐ Whether the first three public features are the right three (proposed: alive map, Reveal Ladder with private voice, culture calendar)
- ☐ Monthly spending limit for the budget breaker (the most you are willing to pay in a month)
- ☐ Invite-only waves or an open launch
- ☐ Which fallback realtime host to keep ready (proposed: Colyseus)

Corrections made since the earlier drafts. Voice moved from LiveKit to Cloudflare Realtime because it is far cheaper at small scale. The map moved from hand-drawn to a real-map engine. Banner prices moved below the reported Lagos Life rates. The NSS allowance now uses the real figure. Revision 2 added quality tiers, a risk register and a week-one sprint after a review of lag and cheap-phone risks.

Sources. Citinewsroom, 2023 NSS allowance approval, Citinewsroom, 2026 allowance proposal, Condia on Lagos Life, WeeTracker on Lagos Life, Protomaps, Protomaps FAQ, Overture Maps places guide, Cloudflare Durable Objects pricing, Cloudflare Realtime pricing, Supabase Realtime limits, Google Maps geo guidelines, OpenAerialMap terms